GG-I-446E
June 23, 1978
SUPERSEDING
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March 28, 1968

FEDERAL SPECIFICATION

ILLUMINATOR, X-RAY FILM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers illuminators used in the viewing of medical x-ray films of all sizes up to and including 14 by 17 inch sheets.

1.2 Classification. The illuminators shall be of the following types, classes and sizes as specified (6.2):

Type I - Single illuminator for table or wall surface mounting.
Type II - Single illuminator for wall recess mounting.
Type III - Panoramic illuminator for table or wall surface mounting.
Type IV - Panoramic illuminator for wall recess mounting.
Type V - Panoramic Illuminator on a mobile base.

Class A - Single deck panoramic illuminator.

Size 1 - Two 14 by 17 inch sections side by side.
Size 2 - Three 14 by 17 inch sections side by side.
Size 3 - Four 14 by 17 inch sections side by side.

Class B - Double deck panoramic illuminator.

Size 4 - Four 14 by 17 inch sections, two on lower deck and two on upper deck.
Size 5 - Six 14 by 17 inch sections, three on lower deck and three on upper deck.
Size 6 - Eight 14 by 17 inch sections, four on lower deck and four on upper deck.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

FSC 6525

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Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, PA, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Specifications:

MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-L-10547 - Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American National Standards Institute (ANSI) Standard:

C73.11 - Plugs and Receptacles - General Purpose, 125 volt, 15 amperes, 2 pole, 3 wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Underwriters' Laboratories, Inc. (UL) Standard:

UL 57 - Electric Lighting Fixtures.
UL 153 - Portable Electric Lamps.
UL 544 - Electrical Medical and Dental Equipment.
UL 935 - Fluorescent Lamp Ballasts.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Road, Northbrook, IL 60062, 1285 Walt Whitman Road, Melville, LI, NY 11746, 1655 Scott Blvd., Santa Clara, CA 95050.)

American Iron and Steel Institute (AISI):

AISI Steel Products Manual - "Stainless and Heat Resistant Steels".

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th St., N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standards for Safety No. 57 and No. 153. Proof shall consist of one of the following:

A. U.L. label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the U.L. report directly to the Contracting Officer, upon the latter's request.

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B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with U.L. standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components,types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.3 Materials.

3.3.1 Tolerances. Standard commercial tolerances shall apply to the thickness of all metals.

3.3.2 Corrosion-resisting steel. Corrosion-resisting steel shall conform to the AISI type 300 series, as indicated in the AISI Steel Products Manual.

3.3.3 Electrical materials and parts. All electrical materials and parts used in the fabrication of the illuminators shall meet the requirements of the Underwriters' Laboratories(UL).

3.3.4 White translucent plastic. The white translucent plastic shall be .125, + .020, - .030, inch thick. It shall have a smooth glossy surface on at least one side and shall be free from bubbles, variations in thickness and flaws in the finish. The light transmission factor of the plastic shall be between fifty and seventy percent (see 6.5).

3.4 Design and construction. Each illuminator shall consist of the following:

a. Housing.
b. Light producing equipment.
c. Viewing panel assembly.
d. Mountings and accessories.

3.4.1 Housing. The housing shall be constructed of either carbon steel sheet or molded fibrous glass-reinforced plastic. The carbon steel shall be at least .032 inch thick (No. 21 Manufacturers' Standard Gage). The housing, combined with the viewing panel, shall provide a rigid enclosure for the fluorescent lamps and the electrical accessories. The housing shall have at least one reflector designed and positioned to distribute light on the surface of the viewing panel in conformance with 3.9. Reflectors for the carbon steel housing shall be constructed of carbon steel sheet and securely mounted to the housing. Reflectors

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for the plastic housing shall be molded integrally with the housing. The housing shall be so formed that the viewing panel may be easily removed to provide access to the lamps and the other electrical components. The housing, as well as the seat for the viewing panel, shall be so constructed that light escapes only by passing through the viewing panel. Louvres, located on the rear of the housing for heat dissipation, shall be acceptable for type I, type III and type V illuminators. If louvres are used, only the seat for the viewing panel need be constructed so that light can escape only by passing through the viewing panel.

3.4.1.1 Type I and type III. The back of type I and type III illuminator housings shall be equipped with keyholes, slots, hooks, or clips to permit mounting on a flat wall surface. Light escaping through keyholes and slots shall be acceptable. Type I illuminators shall be 15-1/2 ± 1/2 inches wide and 19 ± 1 inches high.

3.4.1.2 Type II and type IV. Type II and type IV illuminators shall be designed for wall-recess mounting (3.4.4.2). Each type II and type IV illuminator shall have a flange, flush with the rim of the viewing panel and extending at least 3/4 inch on all four sides, to cover the recess opening. The depth of the housing for type II and type IV illuminators shall not exceed 3-1/4 inches. The type II illuminator housing shall not exceed 18-1/4 inches in height nor 15-3/4 inches in width, exclusive of the flange. The maximum height and width of type IV illuminator housings, exclusive of the flange, shall be as listed in table I.

TABLE I. Maximum height and width of type IV illuminator housings (in inches).

Illuminator	Height	Width
Class A		
Size 1	18-1/4	29-3/4
Size 2	18-1/4	43-3/4
Size 3	18-1/4	57-3/4
Class B		
Size 4	38-3/8	29-3/4
Size 5	38-3/8	43-3/4
Size 6	38-3/8	57-3/4

3.4.1.3 Baffles. All panoramic illuminator housings shall be equipped with light baffles which divide the illuminator into 14 by 17 inch sections. The baffles shall block light from adjoining sections, but they shall not be visible from the front of the viewing panel when they are between two illuminated sections.

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3.4.2 Light producing equipment. The light producing equipment supplied with the illuminators shall consist of the following:

a. Fluorescent lamps.
b. Switch.
c. Ballast.
d. Cord and plug (types I, III and V only).

3.4.2.1 Fluorescent lamps. Each single illuminator (type I and type II) and each section of panoramic illuminators (type III, type IV, and type V) shall be furnished with either two 15-watt daylight fluorescent lamps or one 32-watt daylight fluorescent circline lamp.

3.4.2.2 Switches. Each single illuminator and each section of panoramic illuminators shall be controlled separately by a switch.

3.4.2.2.1 Type I and type II. The switch for the type I and type II illuminators shall be mounted on the front of the illuminator to the right of the viewing panel.

3.4.2.2.2 Type III, type IV and type V. The switches for the type III, type IV and type V illuminators shall be ganged and shall be located at either the lower right front or lower center front of the illuminator.

3.4.2.3 Ballast. Each single illuminator and each section of panoramic illuminators shall contain a ballast suitable for the fluorescent lamps supplied. The ballast shall be class P thermally protected conforming to UL No. 935. The protector shall be the automatic-resetting (thermostatic) type. The ballast shall be the rapid or trigger-start type for operation without the use of a fluorescent lamp starter and suitable for operation on a 110 to 125 volt, 60 Hz line. The rated line current and rated power factor of the ballast shall be such that the illuminator conforms to the maximum temperature rise specified in 3.5.

3.4.2.4 Cord and plug. Type I, type III and type V illuminators shall be equipped with a suitable three-conductor line cord assembled with a Hospital Grade, grounded, three-prong plug as specified in UL 544, which conforms to ANSI C73.11. The exposed portion of the cord shall be at least 7 feet long for type I and type III illuminators and at least 10 feet long for type V illuminators. Type II and type IV illuminators shall be provided with power supply connections suitable for permanent wall-recess mounting.

3.4.3 Viewing panel assembly. The viewing-panel assembly shall consist of the following:

a. White translucent viewing panel.
b. Film-holding device.

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3.4.3.1 White translucent viewing panel. Each single illuminator and each deck of panoramic illuminators shall have a viewing panel consisting of a single sheet of white translucent plastic. The size of the panel for type I and type II illuminators shall be 14 x 17 inches. The width of the panel for class A and for each deck of class B panoramic illuminators shall be 28 inches for size 1 and size 4; 42 inches for size 2 and size 5; and 56 inches for size 3 and size 6. The height shall be 17 inches. The viewing panels for the two decks of the class B illuminators shall be interchangeable. The tolerance on all viewing panel dimensions shall be plus or minus 1/16 inch. When the viewing panel is in place, the smooth, glossy side shall be exposed and the illuminated area for all single illuminators and for each section of panoramic illuminators shall not be less than 13-3/8 by 16-3/8 inches.

3.4.3.2 Film holding device. A film holding device shall extend across top of each single illuminator and each section of panoramic illuminators. The device shall be of the continuous ball (roller-gravity) or strip-spring type. All metal parts of this device shall be fabricated from corrosion-resisting steel. This device shall be an assembly permitting easy removal of the viewing panel or the film holding device, without danger of losing the balls or rollers. The device shall permit easy insertion and removal of any medical X-Ray film at any point across the full width of the panel, without scratching the film. When the viewing panel is vertical, the film holding device shall be capable of holding any standard size medical X-Ray film in place by clamping on one half of the narrow edge of the film. Alternately, the film holding mechanism shall consist of individual spring loaded clips on which rubber composition rollers are mounted to form the film contact surfaces. The spring tension shall provide adequate retaining pressure to hold any standard size medical X-Ray film securely in viewing position by clamping on one half of the narrow edge of the film with the illuminator in the vertical position. The tension shall not be of such value as to hamper the easy insertion and removal of films, or cause film damage. A minimum of four clips shall be equally spaced along the top of each single illuminator and each section of panoramic illuminators. The clips may be hinged to extend forward and serve as wet film holders, supporting a 14 by 17 inch film on its development hanger apart from the viewing panel. The metal parts of this device shall be constructed of corrosion-resisting steel. The clips shall not interfere with the removal of the viewing panel for interior servicing.

3.4.4 Mounting and accessories.

3.4.4.1 Table or wall mounting (Type I and type III). In addition to the keyholes, slots, hooks, or clips (3.4.1.1) to accommodate wall mounting, type I and type III illuminators shall be equipped for table mounting. If required for stability, the illuminators shall be fitted with metal legs or brackets for holding the illuminator so that the viewing panel is vertical when the illuminator is placed on a table or other flat horizontal surface. The points of contact with the supporting surface shall be fitted with rubber pads. The legs or brackets shall be removable to permit the illuminator to be mounted on a wall.

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3.4.4.2 Wall recess mounting (Type II and type IV). All type II and type IV illuminators shall be provided with the necessary fittings for wall recess mounting (3.4.1.2). A flange shall be provided to cover the rough opening in the wall. The illuminators, including flange, when properly installed, shall not offer a horizontal surface upon which dust may accumulate.

3.4.4.3 Mobile base mounted (Type V). Type V illuminators shall be mounted on a mobile base which consists of the following:

a. End supports.
b. Casters.
c. Shelves.
d. Drawer - as specified (6.2).
e. Mounting assembly.

3.4.4.3.1 End supports. The mobile base shall include two end supports of adequate strength and rigidity (3.6) to support the illuminator and shelves.

3.4.4.3.2 Casters. The mobile base shall be supplied with four 3-inch, ball bearing swivel, rubber-tired casters. Wheel bearings shall be ball bearings or self-lubricating, sintered bronze or iron saturated with lubricating oil. The two front casters shall be supplied with brakes. The centers of the casters shall be at least 18 inches apart, front to back.

3.4.4.3.3 Shelves. The mobile base shall be equipped with two shelves securely attached to the end supports (3.4.4.3.1) at heights convenient for film storage. Each shelf shall be at least 16 inches wide and constructed of 0.047 inch thick (No. 18 Manufacturers' Standard Gage) steel sheet.

3.4.4.3.4 Drawer. When specified (see 6.2), the mobile base shall be supplied with a sheet metal drawer supported on the right hand underside of the upper shelf, by slide channels. The drawer shall be at least 20 inches wide, 15 inches deep and 4 inches high and shall have an easily releasable catch.

3.4.4.3.5 Mounting assembly. A mounting assembly shall be provided to mount the illuminator on the base. The mounting assembly in conjunction with the rest of the mobile base shall be so designed that the lower edge of a class A illuminator housing is not less than 48 inches from the floor and the top edge of either a class A or class B illuminator housing is not more than 76 inches from the floor. Class A illuminators shall be pivot mounted to permit rotation of the illuminator from the vertical to the horizontal (viewing panel facing up) position. Means to lock the illuminator securely into position at any angle within the 90 degree range of rotation, without the use of tools, shall be provided. On class B illuminators, unless angulation of both decks is specified (6.2), the top deck shall be held in the vertical position and the bottom deck shall be pivot mounted to permit rotation from the vertical to the horizontal (viewing panel facing up) position. Means to lock the bottom deck securely in the vertical, horizontal and 45^{0} off-vertical positions, without the use of tools, shall be provided.

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3.5 Temperature rise. The illuminator shall be so constructed that when tested as specified in 4.4.2, the average temperature rise on the surface of the viewing panel shall not exceed 15°F above ambient temperature after operating for one hour.

3.6 Strength and rigidity. The mobile base shall be sufficiently sturdy to pass the strength and rigidity test described in 4.4.3 without permanent deformation.

3.7 Dielectric withstand. Each illuminator shall show no sign of dielectric breakdown when tested as specified in 4.4.4.

3.8 Low voltage. The illuminator shall operate properly when tested as specified in 4.4.5.

3.9 Illumination. The maximum brightness of the viewing panel shall be located within a 6-inch square at the center of the viewing panel. No measurement within this 6-inch square shall be less than 450 foot-lamberts. When tested as specified in 4.4.6, the minimum panel brightness shall be no less than 65 percent of the maximum brightness. The areas of the viewing panel within 1-1/2 inches of the edges shall not be included in the brightness measurements.

3.10 Identification marking. Each illuminator shall be marked permanently and legibly with the following:

Contract or purchase order number.
Name or registered trademark of the manufacturer.
Manufacturer's model number.
National Stock Number.
Electrical characteristics.

3.11 Additional marking. In addition, each illuminator shall be marked permanently and legibly, in a conspicuous place, with the following information:

"WARNING - Not to be used below the 5-foot level in Operating Rooms."

3.12 Finish. The exterior of the illuminator, except for plastics and corrosion-resisting steel, shall be finished with a good grade lacquer or baked-on enamel. The color shall be commercial silver-gray hammertone and the surfaces shall be cleaned and degreased and made rust-resistant prior to painting. The interior surfaces of the housing which reflect on the screen shall have a white finish which is highly resistant to color-change, flaking, and chipping. Other interior surfaces shall be made corrosion-resistant by suitable paint, lacquer, or other coating. The illuminator shall be free of chipped, cracked, scratched, and rough areas. All corrosion-resisting steel parts shall have a No. 4 commercial finish and all plastics shall have a clean, bright, smooth finish.

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3.13 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, maintenance, and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.14 Workmanship. Workmanship shall be first class throughout. The x-ray film illuminators shall be free from defects which detract from their appearance or may impair their serviceability. All electrical connections shall be properly made and adequately insulated, so that no bare wires or strands of wire can work loose and come in contact with metal parts, causing a short circuit. All burrs and sharp edges shall be removed.

3.14.1 Welding. Welding shall be durable, sound in every detail and in accordance with the latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal or electrodes where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the illuminator when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturers of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government and provided also that the test methods and the results obtained comply with those described in this specification. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be one illuminator.

TABLE II. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, for an AQL of 1.0. The unit of product for sampling purposes shall be one illuminator.

TABLE III. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Temperature rise	3.5	4.4.2	First article only
Strength and rigidity	3.6	4.4.3	First article only
Dielectric withstand	3.7	4.4.4	100%
Low voltage	3.8	4.4.5	100%
Illumination	3.9	4.4.6	S-2

4.2.3 Raw materials. Each lot of each metal used in the manufacture of the illuminators shall be selected for chemical analysis as specified in 4.4.1.

4.3 Examination. The illuminators shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table IV. Examination shall not be restricted to the classified possible defects listed below.

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TABLE IV. Classification of defects.

Categories	Defects
Major	
101	Any component part missing, broken or otherwise unusable.
102	Exposed edge not free of burrs or sharpness.
103	Viewing panels not free from bubbles, variations in thickness or flaws in the finish.
104	Reflector not as specified.
105	Any component part loose, not secured in place.
106	Warning not complete, permanent and legible.
107	Front surface of viewing panel not smooth.
108	Plug not hospital grade.
Minor	
201	Identification marking not complete, permanent and legible.
202	Finish not as specified or not free of chipped, cracked, scratched, or rough areas.

4.3.2 Dimensional examination. The illuminators shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE V. Examination of preparation for delivery.

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

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4.3.3.1 Examination of closure, waterproofing and banding of container. When the container is required to comply with PPP-B-585, PPP-B-601, PPP-B-621, or PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the appropriate box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition. Four ounces from each lot of material shall be tested in accordance with Fed. Test Method Std. 151.

4.4.2 Temperature rise. Five thermometers or thermocouples shall be fastened to the outside surface of the viewing panel by means of putty or glued cotton pads. One thermometer or thermocouple shall be located at the center of the viewing panel and the remaining four at the corners, 2 inches from each edge. The test shall be started with the viewing panel at room temperature. The illuminator shall be operated at 125 volts for one hour, after which the temperatures indicated by the thermometers or thermocouples shall be noted. The average of the five readings shall be computed to determine the temperature rise. When thermocouples are used, they shall be suitable for the temperature range involved and they shall be used in conjunction with a suitable potentiometer-type indicating instrument.

4.4.3 Strength and rigidity of mobile base (type V). With the illuminator replaced by a rigid piece of wood fastened to the illuminator supports, a vertical load of 225 pounds, minus the weight of the wood, shall be applied to the center of the piece of wood. A 225 pound weight covering an area of one square foot shall then be applied to the center of each of the shelves. The total load of 675 pounds shall remain in place for one hour.

4.4.4 Dielectric withstand. The illuminator shall be tested for dielectric withstand as specified in UL 153.

4.4.5 Low voltage. The illuminator shall be supplied with line voltage of 110-110.5 volts, 60 Hz. The illuminator shall then be checked for proper operation at this voltage.

4.4.6 Illumination. The illuminator shall be operated continuously for a minimum of 50 hours, prior to testing, at 120 volts, 60 Hz. There shall be no ambient light, aside from the illuminator. Lines shall be drawn 1-1/2 inches inside each edge of the viewing panel (types I and II) or 1-1/2 inches inside each section of the viewing panel (types III, IV, and V). The remaining area of the panel - or, of each section - shall be divided into 35 equal rectangles (5 horizontally and 7 vertically). A measurement of brightness, in foot-lamberts, shall be taken in each rectangle, using a photometer (foot-lambert meter), color and cosine corrected, capable of measuring from 0 to 1,000 foot lamberts with a maximum error of 5 percent (see 6.6).

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging.

5.1.1.1.1 Type I thru IV. Each illuminator along with accessories shall be individually wrapped in kraft paper, polyethylene bags, or other suitable non-corrosive material and packaged in a box conforming to PPP-B-636, class domestic. The illuminator and accessories shall be cushioned within the box with fiberboard pads, die-cut polystyrene or other suitable material in such a manner to prevent shifting of or damage to the contents. Closure of the box shall be as specified in the appendix of PPP-B-636.

5.1.1.1.2 Type V. The illuminator shall be disassembled from the illuminator stand. Fluorescent lamps may remain in place in the unit. The lamps in the unit or removed from the unit shall be adequately protected and cushioned against damage during shipping and handling. The illuminator and stand shall be packaged in separate boxes conforming to PPP-B-636, class domestic. Accessories (back support brackets) shall be wrapped in kraft paper or other suitable material and packaged in the illuminator stand box. The illuminator, illuminator stand and accessories shall be blocked, braced and cushioned within their respective boxes with fiberboard pads or other suitable material in such a manner to prevent movement and damage to the contents during shipment and handling.

5.1.2 Level C. Each illuminator of the type specified shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Two unit packages containing type I illuminators or each unit of types II thru IV packaged as specified in 5.1.1.1.1 or each unit of type V packaged as specified in 5.1.1.1.2 shall be packed in a shipping container conforming to PPP-B-585, class 3 style 3; PPP-B-601 overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and banding shall be as specified in the appendix of the applicable specification. Fiberboard box is not authorized as a shipping container for type V illuminators.

5.2.1.1 Case liner. Each level A wood box shall be lined with a case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the appendix of MIL-L-10547. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Two unit packages containing type I illuminators shall be packed in a shipping container conforming to PPP-B-636 class domestic. Closure shall be as specified in the appendix of the box specification. Type II thru type IV illuminators packaged as specified in 5.1.1.1.1 shall require no further packing. Each type V illuminator packaged as specified in 5.1.1.1.2 shall be packed in a shipping container conforming to PPP-B-585, class 1, style 3; PPP-B-601 domestic type, or PPP-B-621, class 1. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged illuminators shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Unit packages and shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The x-ray film illuminators covered by this specification are intended for use in medical facilities. Replacement lamps which are suitable for use with the illuminators are: National Stock Number 6240-00-152-2992 for 15-watt lamps and 6240-00-913-3797 for 32-watt circline lamps.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).
d. Type, class and size required (see 1.2).
e. When a drawer is required (see 3.4.4.3.4).
f. When angulation of both decks is required (see 3.4.4.3.5).

6.3 First article. When a first article is required, it shall be tested under the appropriate provisions of the Defense Acquisition Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

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6.4 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Type	Class	Size
6525-00-604-0000	Illuminator, X-ray film; wall mounted or free-standing.	I	-	-
6525-00-604-0125	Illuminator, X-ray film; mobile base mounted.	V	A	3

6.5 A white translucent plastic which has been found acceptable for the viewing panel is Rohm and Haas Catalog No. W2447, P-80, Plexiglass.

6.6 Photometers which are capable of performing the required measurements include, but are not restricted to, the following:

Photo Research Div., Kollomorgen Corp.	Model PC-200
Vactec, Inc.	Model 3100
Weston Instruments, Inc.	Model 1979

6.7 In this issue, drip trays and hanger hooks (for wet film) have been deleted from the illuminator since there is no longer a requirement for these accessories, except in rare instances. If required, they can be procured directly from commercial sources.

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GG-I-446E ILLUMINATOR, X-RAY FILM

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